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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Cyber Security (CYBER).

Modal verbs terminology

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1 Scope

The present document provides requirements for implementing Human-to-Human Online Preventative Security (H2H-OPS), which complements ETSI TR 104 137 [i.1] 'Human-to-Human Online Preventative Security'. The principles included in the present document are secure by design, safety by design, and privacy by design. These are supported by annexes covering balancing the different requirements for Human-to-Human Online Preventative Security, demography, education and digital literacy.

H2H-OPS is targeted at the operators, designers and service provides that have products, services and applications that allow users to post content online or to interact with each other. For example, social media networks, consumer file cloud storage and sharing sites, video-sharing platforms, online forums, dating services, video games with player-to-player messaging, and online instant messaging services.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ETSI TR 104 137 (V1.1.1) (2026-03): "Cyber Security (CYBER); Human-to-Human Online Preventative Security".

[i.2] NIST SP 800-160 Vol. 1 Rev. 1: "Engineering Trustworthy Secure Systems".

[i.3] NIST Cybersecurity Framework.

[i.4] Government Security Group: "Secure by Design Principles".

[i.5] Ministry of Defence; Secure by Design; Design for security from the start.

[i.6] ISO/IEC 27000 series: "Information security management".

[i.7] ETSI TS 103 645 (V3.1.1) (2024-01): "CYBER; Cyber Security for Consumer Internet of Things: Baseline Requirements".

[i.8] ETSI EN 303 645 (V3.1.3) (2024-09): "CYBER; Cyber Security for Consumer Internet of Things: Baseline Requirements".

[i.9] CISA: "Shifting the Balance of Cybersecurity Risk: Principles and Approaches for Secure by Design Software".

[i.10] Department for Science, Innovation and Technology and Department for Digital, Culture, Media & Sport; 2021: "Principles of safer online platform design; Understand how preventative design measures can reduce the risk of harms happening on your online platform".

[i.11] GDPR: "Privacy by Design".

[i.12] iapp.org: "What is Privacy?".

[i.13] UK Information Commissioner's Office (ICO) "Data protection by design and default".

[i.14] "Digital Regulation Cooperation Forum; 2022: "Online safety and data protection: A joint statement by Ofcom and the Information Commissioner's Office".

[i.15] Open Technology Institute; 2024: "Age Verification: The Complicated Effort to Protect Youth Online".

[i.16] CNIL: "Online age verification: a complex issue with significant privacy risks".

[i.17] GOV.UK: "Users' account details and activity visible to others; improve the safety of your online platform".

[i.18] GOV.UK: "Search functionality; improve the safety of your online platform".

[i.19] Using digital at primary school; internet matters.org.

[i.20] NSPCC Learning: "Online harms: protecting children and young people".

[i.21] Moving to secondary school; internetmatters.org.

[i.22] Student Generative AI Survey 2025.

[i.23] Using AI in the workplace: Opportunities, risks and policy responses; 2024; OECD.

[i.24] NCSC: "Stay safe online top tips for staff infographic".

[i.25] NCSC: "Training; Top Tips For Staff".

[i.26] CLEMi: "Le numérique en famille!; Outils et ressources pour les parents et les professionnels".

[i.27] ageUK: "Staying safe online".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

demography: study of people in a particular area

digital literacy: confident and critical use of a full range of digital technologies for information, communication and basic problem-solving in all aspects of life

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

2FA	two-Factor Authentication
AI	Artificial Intelligence
CISA	Cybersecurity and Infrastructure Security Agency
CISO	Chief Information Security Officer
CLEMi	Centre pour L'Éducation aux Medias et à information
CV	Curriculum Vitae
DoB	Date of Birth
H2H-OPS	Human-to-Human Online Preventive Security
ICT	Information and Communications Technology
ID	Identity Document
IEC	International Electrotechnical Commission
IoT	Internet of Things
ISO	International Organization for Standardization
IT	Information Technology
LGBT+	Lesbian, Gay, Bisexual, and Transgender

NOTE: With the plus (+) signifying inclusivity for other sexual orientations and gender identities like Queer, Questioning, Intersex, Asexual, and others, representing a diverse community whose members are not heterosexual or cisgender.

MoD	Ministry of Defence
NCSC	National Cyber Security Centre
NIST	National Institute of Standards and Technology
PII	Personal Identifiable Information
SP	Special Publication
UK	United Kingdom
USA	United States of America
VPN	Virtual Private Networks
WWW	World Wide Web
ZTA	Zero Trust Architecture

4 Overview of Human-to-Human Online Preventative Security

4.1 Introduction

In the present document, Human-to-Human Online Preventive Security (H2H-OPS) by design is a proactive approach to security and safety that integrates threat identification, risk assessment, and security controls directly into the design and development process of a service and/or a product [i.1]. This includes putting user safety and rights at the centre of the design and development of online products and services. This requires applying the appropriate technical and organizational measures to implement, for example, the data protection principles effectively and safeguard individual rights. Secure by design is the foundation that integrates safety and privacy design into the product or service. However, the type of product or service being provided or used by the user will affect the balance between safety measures and privacy measures. This requires risk assessments and depends on what type of user is expected. For example, if children are going to use it or it has user-to-user messaging that requires additional measures. But if it is not likely to be used by children and messaging is not part of the product, then privacy measures can be increased. Because there are situations where implementing certain types of safety measures requires collecting additional information about users.

4.2 Purpose and Implementation

A significantly diverse array of analyses and legislative findings makes plain that social media networks, messaging applications, and associated support services, including search engines, give rise to a broad array of harmful actions and attacks on individual users, institutions, and society, including fundamental human rights and values. These developments and actions constitute a form of serious cybersecurity threat. This is not a new problem, as the issues began to emerge when the first social networks arrived in the form of Dial-in Bulletin Board Systems. However, the rapid evolution of social networks and associated services today, especially when combined with manipulative Social Network algorithms and Artificial Intelligence, profoundly increases the potential harm and attacks.

When treating these issues, there is a tendency to think of social media as a single service or application. The present document provides a holistic approach to understanding the relations between users, harms and the platforms they use, similar to understanding the concept of an attack surface to improve cybersecurity. In the present document, the Human-to-Human Relationship is established as an interactive process. The inaugural meeting or encounter may immediately establish a connection. Unfortunately, this connection may not be positive. Through the emergence of various personal identities, both humans attempt to relate to or find meaning in their encounters.

There are three parts of H2H-OPS, in clause 5 is Secure by Design, clause 6 is Safety by Design and clause 7 is Privacy by Design.

5 Secure by Design

5.1 Introduction

Secure by design is a principle of cybersecurity and systems engineering that requires systems to be built with security as a foundational property rather than as an afterthought. It is concerned with embedding protections at the earliest design stages of hardware, software, and services, so that security requirements shape the architecture itself, rather than being retrofitted later through patching or external controls. This clause provides an overview of the principles, and it is worth noting that they overlap and complement other design paradigms, such as Zero Trust Architecture (ZTA), privacy by design, and safety by design. There are various standards and guidelines available for secure by design, including but not limited to:

- National standards such as the USA's NIST SP 800-160 (Systems Security Engineering) [i.2] and the NIST Cybersecurity Framework [i.3].
- Government policies, including the UK's Secure by Design Policy for digital services [i.4] and the UK's MoD's 10 Secure by Design principles [i.5].
- International standards, including the ISO/IEC 27000 series [i.6] and ETSI TS 103 645 [i.7] / ETSI EN 303 645 [i.8] (IoT security).
- International Guidance, including the USA's CISA Secure by Design documentation [i.9].

5.2 Principles

The following design principles are relevant to the present document and apply to designers, organizations and service providers, the operators, which have products that offer user to user interaction and sharing of content online:

1) The operators shall create responsibility for cybersecurity risks. Assign risk owners to be accountable for managing cybersecurity risks for a service throughout its lifecycle. These should be senior stakeholders with the experience, knowledge and authority to lead on security activities.

NOTE 1: This is often the Chief Information Security Officer (CISO) a senior-level executive who oversees an organization's information, cyber, and technology security. The CISO's responsibilities include developing, implementing, and enforcing security policies to protect critical data.

2) The operators shall design usable security controls. This requires performing regular user experience and user interaction research. Afterwards operators should implement findings into the service design to make sure security processes are fit for purpose and easy to understand.

NOTE 2: This should create an outcome of a secure service with security controls that minimize friction for users. While insecure practices are avoided by removing incentives for users to find workarounds.

3) The operators shall take ownership of customer security outcomes. This includes steps such as eliminating default passwords and ensuring that the burden of security is not placed on the end user i.e. the customer.

NOTE 3: The operators instead of expecting customers to patch vulnerabilities or configure complex settings, involves building security directly into the product lifecycle to deliver demonstrably safe and secure results out of the box.

4) The operators shall minimize the attack surface. They should use only the capabilities, software, data and hardware components necessary for a service to function, to mitigate cybersecurity risks while achieving its intended use.

EXAMPLE: An organization could shrink their digital footprint, by eliminating unnecessary software, restricting access points and harden configurations so attackers have fewer paths to exploit.

5) The operators shall embed continuous assurance. This requires implementing continuous security assurance processes to create confidence in the effectiveness of security controls, both at the point of delivery and throughout the operational life of the service.

NOTE 4: This should create an outcome where risk owners are provided with evidence that security controls and capabilities operate as intended. Also, the service is built and maintained with the appropriate controls required to mitigate the security risks along with security controls that operate effectively and are updated to reflect changes in the service or threat landscape.

6 Safety by Design

6.1 Introduction

Safety by design (online) is the process of designing an online platform to reduce the risk of harm to those who use it. Safety by design is preventative [i.10]. It considers user safety throughout the development of a service, rather than in response to harms that have occurred.

By understanding how a platform's services expose users to risk, designers can put in place safety measures to protect users from harm. The platform's users may be at increased risk of online harms if the platform allows them to:

- Interact with each other, such as through chat, comments, liking or tagging.
- Create and share text, images, audio or video (user-generated content).

Safety by design can reduce online harms by preventing them before they happen. Platforms can do this by taking a safety by design approach, which can benefit their business or organization by:

- Creating an environment where users feel safe and make safer choices.
- Preventing problems that might be difficult or costly to solve later on.

It is not foolproof and ideally should be regularly reviewed and updated when new risks are identified or incidents occur.

6.2 Principles

1) Users shall not be left to manage their own safety. Anyone who owns or manages a platform where users can interact with each other should take preventive steps to make sure their service reduces a user's exposure to harm.

For example:

- makes a user aware when they do something that might harm themselves or others;

- helps a user report content or behaviour that they think is harmful;
- makes it harder for users to upload or share content that is illegal or breaks a platforms term of service.

2) Platforms shall consider all types of users. Many factors can increase a user's risk of being a victim of harm. Platforms should aim to understand the people who use their service so they can be aware of the risks their service might present to them. This requires an inclusive design that considers the needs of all users.

NOTE 1: Platforms should consider users:

- with protected characteristics which may make them victims of discrimination, for example, race, disability or sexual orientation;
- with low levels of media literacy, which can be a result of age, background or level of education;
- of different levels of ability, for example, children, or learning-disabled people;
- who have accessibility needs, for example, visually impaired people;
- who cannot speak or speak a limited amount of the primary language of the platform.

3) Users shall be empowered to make safer decisions. Platforms should give users the tools and information they need to make safer choices online. They can do this by designing their platform to promote safer decisions.

For example:

- warning users about excessive screen time;
- prompting users to review their privacy settings on a regular basis;
- highlighting when content has been fact-checked by an official or trustworthy source.

NOTE 2: The platform design does not limit a user's ability to make informed choices. For example, using algorithms to recommend content that is harmful to a user, over which they have no or limited control. Ideally, a platform design helps users understand the reliability and accuracy of the content they are interacting with and how their online activity is seen by others, and how to manage that, such as by changing privacy settings or blocking a user.

4) Platforms shall be designed to keep children safe. Children are generally less able to understand risk, which makes them particularly vulnerable to online harms. Even if the online platform is not targeted at under-18s, the platforms should take steps to ensure that only users who are old enough are able to access their service. This could mean undertaking measures to identify child users, using age assurance or verification solutions and tailoring users' online experience in line with their age.

NOTE 3: Designing for children should include:

- Ensure information like terms of service, or tools used to report harms, is prominent and easy to understand for children of different ages.
- Encourage safe, positive interactions and ensure users can interact free from abuse, bullying and harassment.
- Limit access to certain features, functions and content which pose a greater risk of harm to them.
- Give responsible adults (commonly parents, legal guardians, or jobs with safeguarding reasonability's) the ability to shape a child's online experience using safety settings.
- Set safety, security and privacy settings to high by default.
- The platform should establish a clear process for tackling abuse and regularly check that reporting and moderation processes are being enforced effectively.

7 Privacy by Design

7.1 Introduction

The topics of "privacy by design" and "privacy by default" are related to data protection. The term "Privacy by Design" can mean "data protection through technology design" [i.11]. This means there are different ways to achieve "privacy by design", which depends on what is meant by privacy itself. Privacy can be defined as the right to be left alone, or freedom from interference or intrusion. Information privacy is the right to have some control over how personal information is collected and used.

Most people (the end-users or customers), when it comes to privacy, know about it or understand it from conversations about data breaches, wearable tech, social networking, targeted advertising miscues, etc. [i.12]. Also, various cultures have widely differing views on what a person's rights are when it comes to privacy and how it should be regulated.

Privacy and security, while they complement each other, are not the same thing. Data privacy focuses on the use and governance of personal data, for example, by implementing policies to ensure that consumers' personal information is collected, shared, and used in appropriate ways. Security focuses more on protecting data from malicious attacks and the exploitation of stolen data for profit. While security is necessary for protecting data, it is not sufficient for addressing privacy.

There are seven 'foundational principles' of privacy by design [i.13]. Although privacy by design is not necessarily equivalent to data protection by design, these foundational principles can nevertheless underpin any approach that is taken.

7.2 Principles

1) Proactive not reactive; preventative not remedial. A proactive approach shall be taken to data protection, anticipating privacy issues and risks before they occur, rather than waiting until after the fact. This does not just apply in the context of systems design; it involves developing a culture of 'privacy awareness' across an organization.

2) Privacy as the default setting. When designing any system, service, product, or business practice, it shall protect personal data automatically. With privacy built into the system, the individual does not have to take any steps to protect their data; their privacy remains intact without them having to do anything.

3) Privacy embedded into design. The system, operators and platform designers shall embed data protection into the design of any systems, services, products and business practices. It should ensure data protection forms part of the core functions of any system or service, meaning it becomes integral to these systems and services.

4) Full functionality - positive sum, not zero sum. The system, operators and platform designers shall ensure compliance with all legal and regulatory obligations when incorporating all legitimate objectives into their product or service. Also referred to as 'win-win', this principle is essentially about avoiding trade-offs, such as the belief that in any system or service, it is only possible to have privacy or security, not privacy and security.

5) End-to-end security - full lifecycle protection. The system, operators and platform designers shall put into place strong security measures from the beginning and extending this security throughout the 'data lifecycle', i.e. process the data securely and then destroy it securely when it is no longer needed.

6) Visibility and transparency - keep it open. The system, operators and platform designers shall ensure that whatever business practice or technology is used operates according to its premises and objectives and is independently verifiable. It is also about ensuring visibility and transparency to individuals, such as making sure they know what data is processed and for what purpose(s) it is processed.

EXAMPLE: Terms of services can be written to be clear and easy to understand by the users. These measures include but not limited to using plain language this means the text explains key information to someone with no prior knowledge is able to understand; be specific and not vague, this means to be clear about what is in scope and what is not; organize with headings and bullet points, this allows the easy finding of crucial clauses; update regularly, to ensure the terms reflect the latest laws and feedback from user interactions.

7) Respect for user privacy - keep it user-centric. The system, operators and platform designers shall keep the interests of individuals paramount in the design and implementation of any system or service, e.g. by offering strong privacy defaults, providing individuals with controls, and ensuring that appropriate notice is given.

Annex A (informative): Balancing the different Requirements for H2H-OPS

A.1 Introduction

This annex discusses the balancing of the different requirements of online preventative security by design. The foundation required for online preventative security by design is making use of secure by design principles for a product or service. When it comes to privacy and safety, while some measures for them are complementary, other measures can be in opposition. Privacy is intended to grant individuals autonomy over their personal information against unwarranted surveillance and intrusion into personal lives. It should establish boundaries that protect against overreach. Privacy is vital for building trust in the digital landscape, which fosters confidence in online engagements such as e-commerce or social media, where users rely on the assurance that their privacy will be respected.

However, there is a balancing act between safety and privacy in social media. A social media platform utilizes algorithms to combat online harassment and cyberbullying, aiming to create a safer digital environment. However, this strategy requires ongoing monitoring of user interactions and content. Users may applaud the platform's commitment to safety but be apprehensive about the extent of data collection and the risk of personal information misuse.

Balancing different requirements for Online Preventative Security by Design is about understanding the users' behaviour and how to protect them from harm while also allowing them to participate in online activities.

Different examples of this balancing act are found in the areas of age verification, signing in or up to online services and user participation in online services. These examples are not comprehensive, and other interactions between online safety and privacy will occur.

A.2 Age Assurance

Online age assurance is a complex issue regarding privacy, access to content (often to stop children accessing age-inappropriate content) and data protection [i.15]. Verifying the age of an Internet user is hampered by the difficulty for the various stakeholders on the Internet to really know who the person is behind the computer or smartphone. The need to identify Internet users is an issue for privacy and personal data protection, since knowledge of an individual's identity can then be linked to their online activity. Yet, this contains particularly sensitive, private information.

In order to visit certain sites or for certain online activities, it is by nature necessary to identify oneself (e.g. to buy an age-restricted good like alcohol or cigarettes on an e-commerce site) [i.16]. Age verification, when necessary, takes place in a context where the site publisher already knows certain elements of the identity or personal data of the Internet user (including banking details). On the other hand, where access to the site or an online service does not necessarily require identification, age verification is likely to alter the protection of the Internet user's privacy, by preventing them from visiting the site if they do not provide the publisher with information on their identity.

An age assurance method may be at odds with data minimization, posing risks to user data privacy and security. Best practises require that online operators cannot knowingly retain users' Personal Identifiable Information (PII), but the act of verifying user ages itself can put personal and sensitive data at risk. For instance, operators verifying users' ages through government-issued ID or credit card information put data at risk if secure processes are not in place for use, collection, processing, storage, or deletion of PII. This, in turn, increases the risk that such sensitive data could be merged, stolen, sold, or turned over as part of legal proceedings.

At the same time, operators who choose to verify ages through estimation or inference models may increase surveillance and monitoring of users' online activity, such as their content, engagement, social networks, geographic location, screen time, linked accounts, and browsing history. Subjecting users to such intrusive practices may result in an effect that suppresses online speech and enables the potential collection, use, or sale of user activity data.

Age assurance methods are not foolproof, and nor will they completely stop underage users from intentionally or unintentionally accessing age-inappropriate content. Users can still use tools like Virtual Private Networks (VPNs) to bypass age verification. Also, users evade restrictions through a variety of methods, for example, users can evade age assurance by borrowing the device of a parent or adult or by buying, renting, or trading verified adult accounts.

Further, as the technology has developed, users have been able to use generative AI to circumvent age verification methods. For example, users could use realistic filters that can alter the age a person is perceived as in images and videos, or they could generate an image of an accepted identification document. Or even simply use near photo-realistic character creator tools from video games to fool age assurance methods that use facial age estimation.

A.3 Signing in or Up to Online Services

The type of personal information that users could provide to access platforms varies. Some online services offer access to content without the need to subscribe or create a profile, while others might require users to create a profile or verify certain information about themselves (for example, their name, age and contact email address) before they can access services or specific types of content [i.14]. Also, users might have anonymous or multiple accounts. A user may wish to protect their identity, for example, those in LGBT+ communities, whistleblowers and victims of domestic violence.

There are also valid reasons for creating multiple accounts, such as for work and personal activity. But some users may use these functions to harm others online. For example, if users are able to hide their true identity from others, some may be encouraged to engage in harmful behaviour, for example, by creating multiple accounts, they can contact others even if they have been asked to stop or have had a previous account disabled [i.17]. Harms that can occur as a result of anonymous or multiple accounts include:

- cyberbullying and cyberstalking;
- child sexual exploitation and abuse;
- terrorist content;
- hate crime;
- disinformation.

In general, there are different steps to reduce harm, including but not limited to:

- restricting the ability to reduce safety or privacy levels for child or unverified accounts;
- prompting users when they message an unverified account, asking them to confirm they understand the risks before allowing them to continue;
- prompt users to additionally verify their accounts during account creation, for example, using two-Factor Authentication (2FA).

A.4 User Participation in Online Services

One way in which online service providers present content is by using content recommendation algorithms, or systems which aim to rank or return content that matches a user's perceived interests. A large number of factors are used to do this, including information that constitutes the user's personal data. How algorithms work can lead to a range of different outcomes for users, including directing users to content that they will interact with for longer, or encouraging users to engage with content posted by their friends. But they can also lead to harmful content being amplified.

Where personal data is used to determine the content served to users, data protection law will apply [i.18]. For example, services need to be transparent about the way in which they process personal data to make content recommendations. This extends to data processing in connection with the online safety measures that are incorporated into their content recommendation systems, such as identifying vulnerable users who could be placed at risk by certain types of content.

There are different steps to reduce the risk of harm when users participate in online services, including but not limited to:

- blocking high-risk search terms relating to illegal harms and activity;
- ensuring that auto-suggestion functions and algorithmic recommendations do not lead people to harmful content;
- ensuring that harmful content does not appear at the top of search results;

- providing links to resources and support to users who search for high-risk search terms;
- users' content, contacts and activity are only visible to friends;
- users cannot share their location with strangers;
- automatic face recognition is turned off for photos.

Annex B (informative): Understanding Demography, Education and Digital Literacy

B.1 Introduction

This annex aims to provide a broad understanding of how different people, divided into age ranges, use apps and devices. This is a constantly changing area, and what is true today might not be true tomorrow. Also, this annex does not cover the other factors that can affect how different users use apps and devices, as there are differences for genders/sexes, ethnicities, nationalities, etc. It is also important to note that different groups are affected differently by the harms online.

A key part of reducing and preventing harm is for users to have the knowledge to identify the potential risks and to be conscious of personal security while browsing, sharing or surfing the internet through different apps, services and devices they may use. This includes education that enables children (and parents), akin to teaching children to be street aware, but online. The difficulty is that many adults were not brought up with technology, and their children understand more than they do. It is important for parents to be acutely aware of the dangers posed to children while they are exposed to the internet. Practically all parents understand the dangers of unsupervised youngsters talking to strangers, and/or accepting some presents from them. Parents and teachers across the globe need to realize that nowadays, allowing children to navigate the World Wide Web (WWW) is like placing those children into an unknown geographical location, completely on their own, unsupervised. Parental controls only go so far, but it does not help them as they get older, as children, especially during their teenage years, when children begin to explore and seek greater independence.

One example of teaching adults is that parents also need to understand the risk of providing incorrect Dates of Birth (DoBs) to allow their children access to social media. If a parent provides an incorrect DoB for their child, that child could be placed at greater risk of encountering age-inappropriate or harmful content online. Once a user reaches age 16 or 18, some platforms, for example, introduce certain features and functionalities not available to younger users - such as direct messaging and the ability to see adult content.

The following clauses build on each other. So, ideally, the older a person, the knowledge they have to stay safe and secure in a digital world, which always grows and is reinforced by refreshing core lessons. Additionally, information targeted at one group can be just as relevant for another, regardless of age group. For these age groups, people within them will have different knowledge and confidence levels with technology, which will affect their ability to solve problems and mitigate risks.

NOTE: The following clauses with the different age groups are indicative not absolute.

B.2 Ages 6 and Under

This includes children who are in pre-school and at the beginning of the primary education level. Preschoolers learn by interacting with the world around them. Parents are using digital devices with their young children both to support learning and to keep them quiet. They are also using the internet and social media to get ideas for learning and play activities. If very young children are supported by adults when they use digital devices, then they soon become able to use them independently for some tasks, including talk and literacy-related activities, such as recognizing and purposefully selecting screen symbols and icons. They develop the technical and motor skills needed to operate touch screens - to swipe, locate, and tap to select and use apps such as the camera and photo gallery to take and view photos and videos, or to access online media. Unsupervised use has a risk of young children being exposed to material they are not developmentally ready for. Or when using a parent's device unsupervised, making accidental unauthorised purchases either via touchscreen apps or voice-controlled smart audio devices.

The age at which digital literacy needs to start is rapidly diminishing. Children, as young as 1 year old, notice their parents' smartphone devices and are interested in pressing the touch-sensitive screens, where displays change rapidly. Before these very young children can read or write, they know how to turn on various applications on these smart hand-held devices and manipulate them. Although at this young age, parents are typically in control of what applications their young child can open and play with on their smart handheld device, it is advisable to start cyber education with children as young as possible.

B.3 Ages 7 to 11

This includes children who are at the primary education level. At this age, most children use apps and websites for entertainment and begin to use devices for schoolwork. Typically, children will have their first user experience with a parent's computer, tablet, smartphone, etc. designed and sized for adult users. At this stage in life, there is a risk that if a child has unsupervised internet access, there is a chance of them being exposed to themes they are not developmentally ready for. Information received by children beyond their experience will be interpreted within the framework of their experience, sometimes causing great anxiety and distress. For example, overexposure to themes such as war and sex may cause great difficulty later on, as children have to reinterpret knowledge [i.19]. Online activities include watching multimedia content on video-sharing platforms, playing video games online, and some are beginning to use social media platforms, even though most have a minimum age of 13. Along with using education apps to supplement school learning.

For this age group, children are becoming aware of themselves and group dynamics [i.20]. This is also an age when they may start to use the internet and access online content unsupervised. They should be taught through lessons and exercises. For example's passwords should not be shared with strangers. To start teaching them to understand not to share personal information online. Also, to be socially responsible, such as online bullying is just as bad as physical bullying.

There is a chance they will see or experience harmful content online either accidentally or intentionally. They should always be offered support. What this entails will depend on the nature of the harmful content, the context of the situation and the individual child or young person's needs. Remember, if a child or young person has actively sought out the content, this may indicate there is another concern in the young person's life. These measures are targeted more at schools and teachers.

Depending on the type of harmful content a child or young person has seen, different steps could be taken, including but not limited to:

1) Help them understand or process what they have seen.
2) Answer questions or allay fears or worries.
3) Make sure they have a safe space to talk in the future, if needed, rather than sharing harmful content with friends.
4) Support them to block or report content to stop them from encountering it again.
5) Consider what steps can help them to stay safe online in the future.
6) Make sure they know who to get help from if they ever see anything that's worrying or upsetting.
7) Provide support for any underlying issues.

B.4 Ages 12 to 18

These are children who are in secondary school, going through their teenage or adolescent years, which brings many changes not only physically, but also mentally and socially. As children transition from primary education to secondary, they experience many digital firsts, with many getting their first smartphone [i.21]. During these years, adolescents increase their ability to think abstractly and eventually make plans and set long-term goals. Each child may progress at a different rate and may have a different view of the world. This can be a highly emotional time. They are coping with hormonal changes brought about by puberty. They are also hitting a time when peers will have the most influence on them. This is a period of life when some kids start dabbling in riskier behaviour (self-harm, smoking, drug use, sex, etc.). This can be exacerbated by access to certain types of internet platforms and services, which can include social media, photo/video sharing platforms and instant messaging services. Teenagers essentially communicate as adults, with increasing maturity throughout their secondary education. As teenagers seek independence from family and establish their own identity, they begin thinking abstractly and become concerned with moral issues. All of this shapes the way they think and communicate. During this time, they will begin to interact and consume social media along with online content, which may influence the development of their identity.

This is an age when a child is becoming a teenager and will seek greater independence in their life and online. This can expose them to risk and harm; therefore, it is important to teach them to understand a range of ways to use technology safely, respectfully, responsibly and securely, including protecting their online identity and privacy; recognizing inappropriate content, contact and conduct; and knowing how to report concerns.

They will be encouraged to talk to their parents and teachers about concerns or incidents they may experience. There is a chance they will feel embarrassed or worried about taking that step. So, they should be aware of resources, for example 'nofiltr.org', which seek to provide a non-judgemental space to get support and to help people learn to be safe online. This allows teenagers to find resources to learn more about the potential risks they may face online, learn how to navigate them, and get help when they need it. Give or seek advice from their peers or a professional, knowing when to block or report someone or identify an online grooming situation. As well as test their knowledge by exploring quizzes around topics like digital boundaries, red flags, online self-care, and their online persona.

During this time, they should learn:

1) The importance of using unique passwords and multifactor authentication. This can also include methods such as passkeys.

2) Understand what personal identifiable information is and how it is used by companies online, and if shared without due consideration, how it can affect them.

3) Understand what social engineering is and how to spot the signs, along with being able to identify an online scam and how to identify what is presented as fact vs fiction online.

4) Understand the dangers in spreading misinformation online, not just in general, but also the harm when it happens to a person.

5) Understand the dangers of downloading files from untrusted sources and avoid clicking on unknown, suspicious links.

B.5 Ages 19 to 29

This age range includes people in tertiary education, who are beginning to start their working life. During this time, people are now legally adults (the age of majority), though the age varies from country to country. In the majority of places, the age is 18, but in some countries, it is younger, while in others, it is older. Also, some countries have different ages within them and are often separate from the legal age to carry out certain activities, such as buying alcohol or gambling, etc. This age range is considered digital natives; they grew up with smartphones and the internet. Generally, they prefer apps that provide visual stimulation, and this group consumes the most video content. Within this age group, but not limited to, there is a trend of increasing use of generative AI tools, with the main reasons being as a way to save themselves time and to improve the quality of their work. Though there is an acknowledgement of the risk of getting false or biased results [i.22].

This is an age where people are considered adults, and they are leaving their parents' home, getting jobs, going to university and are on the road, becoming independent adults. They should be aware of different types of online threats. These include malware, phishing attacks, and scams. Understanding how these threats work helps users to detect and reduce the risk of being affected by them. At this stage in life, they should have been taught different good practices to stay safe and secure. Including but not limited to:

1) Safeguard Passwords: Use a strong password for each account. Enable two-factor authentication when available.

2) Exercise Caution with Suspicious Messages: Be wary of phishing attempts. Avoid clicking links or downloading attachments from unknown sources.

3) Reduce Public Wi-Fi® Usage for Secure Activities: Public Wi-Fi® networks can be tempting to use, but they can also be vulnerable or fraudulent. Limiting their usage for sensitive tasks like e-banking or online shopping is advisable.

4) Practice Discretion in Social Media Sharing: Be mindful of what is shared on social media. Also, knowing how to adjust privacy settings to control the visibility of personal content.

5) Enhance Security with Two-Factor Authentication: Enable two-factor authentication whenever possible. Understand, this step is necessary to add an extra layer of protection to accounts.

6) Maintain Updated Apps and Software: Keep their devices, applications, and software current.

7) Safeguard Their Personal Information: Be cautious about sharing sensitive information online. Provide data only on secure, reputable websites.

8) Shop Smart and Securely Online: Be careful when purchasing online. Ensure the website is trustworthy and uses secure payment methods.

9) Stay Vigilant and Alert: Have a critical mindset, and question suspicious offers/requests. Also, think twice before sharing personal information or engaging in risky virtual activities.

Though this age group is considered to be digital natives, that does not mean they are naturally better at understanding risks and knowing how to stay safe online and problem-solving issues with technology. Part of it is due to how the majority of devices and services have a friendly user experience-focused design, which requires little input from the user to set up and use.

B.6 Ages 30 to 55

This age range is broad and diverse. Generally, people in this age will have their careers and have their own families. Generally, use a wide variety of apps and services for work and personal use. They will be the likeliest to respond to push notifications. They may prefer personal interaction rather than visual stimulation when compared to younger users of apps. They may also have a burden of responsibility of raising their children while at the same caring for older relatives and parents. There is also the rapid change and use by employers to use AI applications to sift through CVs, interact with customers, allocate, direct, and evaluate work, and identify and provide training. Also, workers are using AI in an increasing number of tasks. Depending on the type of work, it is being used to improve productivity and reduce the burden of certain tasks. It is important to note that AI may bring increased work intensity and the collection and use of data, amongst others plus the risks include: bias and discrimination, unequal impact on workers, lack of human oversight, as well as lack of transparency, explainability and accountability, will not just this age groups but all the other age groups as well to different degrees [i.23].

Generally, this is a time when people have careers and families. Often, the companies will be responsible for ensuring their employees receive cybersecurity training, and the depth and knowledge will vary for a job's role and responsibilities. Training resources from organizations such as the UK's NCSC [i.24], [i.25] are good starting places. They give advice about passwords, devices, phishing and reporting:

1) Creating strong passwords:

   - Use three random words. Ideally, trying to make the words as random and unrelated as possible. Avoid using predictable passwords, such as dates, family and pet names.

   - Two-step verification is enabled.

2) Keeping devices secure:

   - Secure a device with a screen lock. This can be a PIN, password, biometric or pattern. A user should pick one of these that they can stick with.

   - Be aware of surroundings: Consider who's around them. Be aware of others around them who might be overlooking their screen or listening in on their conversations. For example, consider using privacy screens, particularly if they are regularly using devices on the move.

3) Defending against phishing:

   - Manage digital footprint: review privacy settings and keep personal information to a minimum.

   - Remember the key identifiers: always check emails for signs of urgency, using the authority of the sender, imitation attempts and spelling or grammatical errors.

   - Know policies and processes: read an organisation's policies and processes to help know what to do.

   - If in doubt, check it out: if a user makes a mistake, or they are unsure about something, then they should always report it within their own organization and forward it to the relevant country's cybersecurity or online fraud organization to investigate.

4) Reporting incidents promptly:

   - Cyberattacks can be difficult to spot, so people should not be hesitant to ask for further guidance or support when something feels suspicious or unusual.

   - Report attacks as soon as possible; do not assume that someone else will do it. Even if they have done something (such as clicked on a bad link), they should not be afraid to always report what's happened.

There are resources available for parents and guardians to help them have dialogues with their children about online risks, privacy and cybersecurity. For example, 'Le Centre pour l'Education aux Medias et à information (CLEMi)' [i.26] has themed guides on cyberbullying, information and disinformation, social networks, screen time and content. They are designed to provide activities to educate and enable media literacy regarding online risks and harm. They cover issues including, but not limited to:

1) Identify, understand, and talk about cyberbullying.

2) How to recognize misinformation and talk about it with children.

3) How to protect personal data.

4) Advice on talking about social media with teenagers.

5) Managing children's screen time.

This age group experienced the analogue-to-digital transition and the development of current internet-connected information technology. This can mean they have better knowledge and experience in problem-solving issues with devices and a better understanding of how to be safe online and the importance of good cybersecurity practices.

B.7 Ages 56 to 75

This age range tends to be nearer the end of their working life and beginning to look towards retirement. They may be more financially secure, for example, their mortgage is paid off, and their children have left home. They generally use apps and services that help maintain a better quality of life, but not all in this age range may be adept at adjusting to new interfaces with the continuing digitalisation of key services necessary to live life, so they are potentially more vulnerable to being left behind as services become harder to access when locked behind a digital wall. Online risks for this age group will differ from others, such as coming up to retirement, they may be targeted in pension scams, while at the same time, they face the same risks of being targeted for romance or sextortion attacks that can affect all ages.

Being online can make life easier for older people in many ways, but it also comes with the increased risk of scams and fraud. Online scams are becoming increasingly targeted at older, potentially less tech-savvy users, but they can protect themselves by knowing what to look out for and what to do if they suspect a scam [i.27]. This includes but is not limited to:

1) Identify scam emails:

   - errors in spelling or grammar, or an unusual style of writing;

   - unofficial or suspicious email addresses or senders;

   - requests for personal information, such as their username, full password or bank details, a genuine organization will never ask for this;

   - threats that unless they act now, a deal will expire, or their account will be closed.

2) Identify a fake website:

   - Check the URL. Does it start with HTTP or HTTPS? Are there spelling mistakes or typos in the name of the company? Is it a recognizable URL? Does it end in .net or .org, as these are rarely used for online commerce sites?

   - Look out for spelling and grammar issues. Watch out for spelling and grammar issues, as well as any odd phrases that may suggest the site was created in a hurry.

- Check if there is a 'Contact us' page. Legitimate businesses and websites will provide a way to get in contact with them.

3) Understand the risks of computer viruses:

- Users may be sent an email with an attachment which, when they click on it, will release a virus.
- Criminals can then use this to take control of their computer, or the virus may scan their computer for personal information. It can also slow their computer down, send out spam emails or delete files.
- They may even get a phone call from someone claiming to be from a well-known software company, saying there is a problem with their computer and that they need to get access to it, including asking for personal details. Legitimate IT companies never contact customers in this way. This is a common phone scam; they should know to hang up straight away.

4) Relationship scams:

- Know that scammers can use social networks or dating websites. Once they have gained a person's trust, they might start asking them for money, often by telling them an emotional story about their life.
- Never send the person money or give them account details. It is always worth talking to a friend or relative about it, especially if things seem to be moving fast. Be wary of the person moving away from the chat room or dating site to communicating by email or text message.
- If arranging to meet, make sure it is in a public place, tell someone else where the meeting is happening and do not give away information too quickly.

Within this age group, there will be a wide range of people with varying levels of knowledge and confidence with the internet and ICT in general.

B.8 Ages 76 and Older

While this age group is increasingly online, there are still people who do not use the internet and do not have a smartphone. Digital technology is playing an increasing role in our lives, and for many people, it is essential to the way that they socialise, work, shop, manage their finances, access services, and get entertainment. For example, smart speakers are gaining popularity with this demographic. They can perform tasks like playing music, reporting the weather, and making calls via simple voice commands, which can be useful for those with mobility issues. However, not everyone is online, while others only use the internet in limited ways. Although many older people fully embrace the digital world, digital exclusion increases with age. There is also a divide in the skills of this age group. So, some may be completely comfortable and know how to stay safe online, for example, recognize suspicious links and know that clicking on these links or downloading unfamiliar attachments is a risk (e.g. spam/phishing emails, texts, pop-ups) others do not have this knowledge or only a limited amount. Most older adults require help setting up and learning new devices. Access to training and support, whether from family, friends, or dedicated programs, is a critical factor for ensuring they are comfortable using smart devices and apps. This age group will also have a higher prevalence of memory disorders, for example, mild cognitive impairment, Alzheimer's. age-related forgetfulness and dementia, etc. This puts them at greater risk of being easy targets for online attacks such as phishing, scams, fraud, and identity theft.

This is an age where the likelihood of cognitive decline increases, which can make people more vulnerable to being scammed or manipulated. A solution to this is for them to organize a mechanism, often a power of attorney, where a trusted (younger) family member, friend or solicitor/attorney has joint control or shared access to their digital assets and/or services. This should be organized before any serious cognitive decline, but it can be done afterwards. This legal process will vary from country to country.

This involves identifying digital assets/services which commonly include:

1) Financial accounts.
2) Email and social media.
3) Digital files: photos, documents stored in cloud services.
4) Online subscriptions.

5) Business accounts.

6) Access to devices.

It is useful to list each asset with login details and provider contact information. This helps to find and manage them when needed quickly.

This can reduce the risk of the person being targeted or affected by scams, identity theft, etc. This should also be done with the consensus of not just the person granting autonomy, but also with other family members, relatives, and friends, as needed, as there is a risk that someone may abuse this for their own gain. Open dialogue about this emotive topic is essential for trust and hopefully prevents misuse.

It should be noted that these issues can also apply to people of any age due to circumstances such as suffering a serious injury or having a cognitive disability, which can leave people at greater risk of being successfully targeted by malicious or bad actors.

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